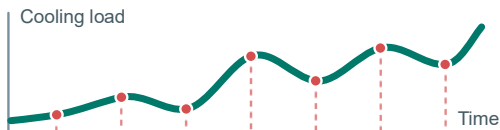


Event-Triggered Predictive Reinforcement Learning with Unsupervised Dynamic Event Gating for Energy Efficient HVAC Control

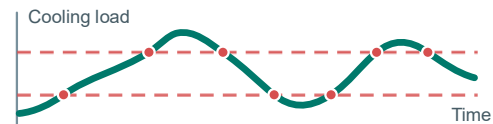
Limitations of conventional triggering

Fixed-interval control



Redundant updates in stable periods

Static-threshold control

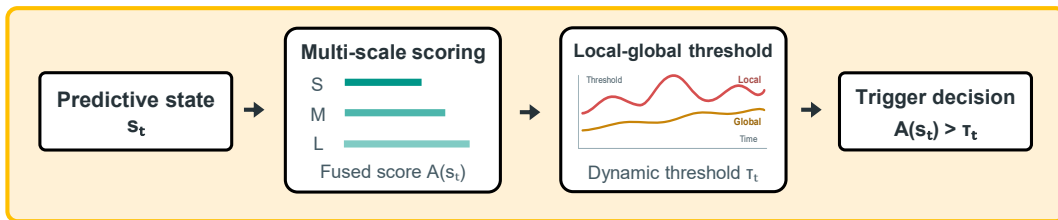


Missed or excessive triggers under drift

Limited adaptability

ET-PRL: Predict, detect, and act on demand

Unsupervised dynamic event gate



On-demand action selection

EVENT DETECTED

Predictive DQN

Update HVAC setpoint

NO EVENT

Hold setpoint

Zero-order hold

Responsive under disturbances Sparse during stable operation

Performance on real chiller data

53.54%

fewer action updates

2.13%

less energy use

Compared with fixed-step RL